

so-heavenly voice, "Let there be chat," and lo, chat was born. IRC (Internet Relay Chat) was the most popular of these at one point in time, but soon gave way to the Yahoos and MSNs. However, what IRC brought along with it was the ability to send a file directly to another user on the same server. Now some may argue that the Usenet BBS (bulletin board service) was the first P2P network, but IRC's @find is what most of us were using to find shared files back in the day.

Then came Napster! The rest, as they say, is popular history. For those of you who haven't heard of Napster's notorious days, or the Metallica vs. Napster battle, cease, desist, do a quick Google, read that, then come back!

The history of P2P and the justice system is made infamous mostly by this loss of a lawsuit by Napster's lawyers. Next, services such as AudioGalaxy, KaZaA, WinMX, eDonkey and the likes were hit. Today, BitTorrent is one of the last remaining bushes in P2P file sharing garden, and it's flourishing! The uploading and downloading on Torrent networks account for a majority of *all* Internet traffic across the world!

Legality

So it's all illegal? Why not just ban it all? Ah! If life were always that simple, the majority of you would have paid for your software, music and movies, and your computer would cost a good deal more...

P2P isn't illegal, per se—if you want to share your vacation video with the world, no one has the right to stop you! If you have a band, make music, and want the world to listen to it, you have every right to do so. The same applies for software you make, any movies you produce, and basically anything that is licensed under Creative Commons or is otherwise open source. There are terabytes of movies, music, applications, operating systems and other digital content that can legally be shared on P2P networks. Granted, legal P2P accounts for almost nothing



A piece of art is not a loaf of bread. When someone steals a loaf of bread from the store, that's it. The loaf of bread is gone. When someone downloads a piece of music, it's just data... Treating your audience like thieves is absurd. Anyone who chooses to listen to our music becomes a collaborator."

Jeff Tweedy
Frontman, Wilco, in an interview with *Wired*
Photo courtesy Plexifilm

of the global P2P bandwidth, but it is this very small segment that P2P software caters to.

No one can blame P2P software for piracy; that shame falls solely upon us users of the software.

Advantages

P2P networks are actually a great boon, especially to open source content creators. Consider this: you are a software developer; you magnanimously decide to create a useful application without wanting to sell it commercially (for example, a Linux distro). You spend months (if not years) developing it, at your own costs, and now find that in order to share it with the world, you will have to incur further costs of buying Web hosting bandwidth. God help your wallet if your software is as popular as say, the Ubuntu Linux distro. With millions of people downloading the 700 MB odd CD image, you're certainly going to get a big bandwidth bill every month!

However, thanks to P2P, you have an option. First you find some friends who have fat pipes at home or work, get them to download the software from you, then make a Torrent, upload it to a popular legal torrent site (or just host it on your own site), and instead of millions of people downloading 700 MB each, you get millions of people downloading a small 7 KB .torrent file each and then downloading the 700 MB from your friends who are also hosting the file. Plus, the more people that finish downloading the file, the more "seeders" you have and the less the strain is on the original hoster's bandwidth! Thus, communities are born! Apart from saving tons of hard cash in bandwidth costs, P2P can help you develop a community, quite like a forum does; and then there are the ideas that don't involve any file sharing at all!

P2P With A Difference

Have you heard of Skype? Again, if you haven't, Google it, read the first link, come back...

Fighting Back

Those of you whose preferred weapon of mass consumption is BitTorrent, using client software such as BitComet, Azureus, etc., might sometimes find that you have downloaded a lot of rubbish data (data that doesn't pass the hash check/checksum). Every torrent is broken up into pieces, and each piece downloaded from many peers in a swarm (collection of peers and seeds that you're connected to). Now each .torrent file contains a hash check/checksum for each piece of the large file, and every time your client finishes a piece, it does a hash check and then compares it to the information in the .torrent file. If the checksum/hash check fails, the data is discarded and has to be re-downloaded.

This can happen for both legal and illegal downloads, because of a network error, garbled data, viruses, and of course, rogue seeders or peers. Recently, however, there have been several complaints of a lot of rubbish data being downloaded by P2P users, especially when downloading pirated content. Upon further investigation by P2P experts, most of said "rubbish data" seemed to be originating from a few companies who seemed intent on foiling the attempts of users to download pirated content. Now rumours on P2P networks suggest that these are companies that have been hired by the RIAA (Recording Industry Association of America) to fight piracy on the

Torrent network, but they just as easily could be virus infected sub-nets. We're not here to judge whether such rumours hold any truth to them, and since quite a few people have reported receiving rubbish data even when downloading legal content such as Linux distros, we've got a solution.

What you need to do is run another program such as PeerGuardian 2 (<http://phoenixlabs.org/pg2/>) along with your Torrent client. PeerGuardian 2 is nothing more than a Windows IP blocker, which prevents you from connecting to any IP that it deems "unsafe." The list of blocked IPs is updated centrally on the Phoenix Labs server by monitoring P2P networks and checking which IPs are sending out rubbish data, for whatever reason. The program on your computer then updates itself with these IPs on a regular basis. Since these IPs are blocked by PeerGuardian 2, Windows does not allow your Torrent client to connect to them, and thus stops or at least reduces the amount of rubbish data you receive.

In order to not block HTTP access to any sites, and thus affect your browsing experience, PeerGuardian 2 allows you to choose what ports to block, and whether to allow access via port 80 (HTTP) to even blocked IPs. It also has some more options of blocking access to US government sites and even sites that record your IP address.